

# Week 1: Conway’s Soldiers

Jumping-peg games — facilitator guide for the three table leaders

One hour; ten children (K–1, grades 2–3, grades 4–5) at three tables; three adults.

**The one rule of the game.** A soldier (a penny) may hop over a soldier on an adjacent square — up, down, left or right, never diagonally — into the *empty* square immediately beyond. The soldier that was hopped over is removed from the board. There are no other moves and no turns: the whole table plays one position together.

## 1 The session at a glance

**The theme.** An army of pennies camps south of a river drawn across a grid. Everything north of the river is empty. How far north can a single soldier get? One row north takes 2 soldiers, two rows take 4, three rows take 8, four rows take 20 — and five rows is impossible no matter how large the army. The impossibility proof (Conway, 1961) prices every square so that no hop can increase the army’s total price, and shows the entire endless camp is worth exactly the price of one square in row 5. The same trick, on a one-row strip, shows a soldier can get one square past the river but never two. All of this is worked out in Section 8; the tables meet it at their own levels.

**Suggested table assignments.** The parent takes the K–1 table (three children, the most hands-on script, no proofs). The second mathematician takes grades 2–3 (four children; the row-3 puzzle and its hint ladder). The organizer takes grades 4–5 (three children; the pricing argument) and also runs the launch and the share, so the 4–5 table starts a couple of minutes late and is fine being left alone briefly at the end. Swap freely; the guide is written so that any adult can run any table.

|                  |                                                                                                                                                      |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>0:00–0:05</b> | <b>Free handling.</b> Pennies and grids are on the tables; children arrange and hop as they like. Adults show a hop when asked but do not teach yet. |
| <b>0:05–0:12</b> | <b>Whole-group launch</b> (organizer, at the checkerboard): the rule, the river, one row north with two soldiers, the question <i>how far north?</i> |
| <b>0:12–0:50</b> | <b>Table work</b> (see the per-table sections). Each table ends by choosing one thing to show.                                                       |
| <b>0:50–1:00</b> | <b>Share:</b> each table shows one thing at the checkerboard; the organizer closes with 2, 4, 8, 20, never.                                          |

|         | K–1 (parent)                                      | Grades 2–3 (mathematician)                            | Grades 4–5 (organizer)                                    |
|---------|---------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------|
| first   | the hop; “can it hop?” pictures                   | the hop; strip puzzles W1–W4 (one impossible)         | strip puzzles S1–S3; <i>how far along a strip?</i>        |
| then    | <i>leave one</i> strips; hops = soldiers – 1      | one row, two rows north; fewest is 4; why not 3       | one, two, three rows north; fewest 2, 4, 8; prove 3 fails |
| then    | across the river (2 soldiers); two rows north (4) | three rows north with 8 (the T army)                  | prices: why square 8 on the strip is impossible           |
| finally | three rows north with the adult; make your own    | explorer’s page: the pattern, other armies, hop count | the 20-soldier army; four rows yes, five never            |

## 2 Materials and preparation

- **Student pages**, printed single-sided: **week-k-1** (6 pages, one copy per child plus one for the adult), **week-grades-2-3** (5 pages, five copies), **week-grades-4-5** (4 pages, four copies). The boards on the pages have 2.0–2.3 cm squares so pennies sit on them directly. Print this guide for each adult.
- **Pennies (or counters)**: about 30 at K–1, 40 at grades 2–3, 60 at grades 4–5 (the 20-soldier army plus spares). Pennies should be all alike; the game has one kind of piece.
- **Printed grid sheets** at every table (squares at least 2 cm). The 4–5 table needs one sheet of at least  $9 \times 8$  squares for the 20-soldier army; draw the river on it in advance along the top of the fourth row from the bottom, with a marker.
- **Launch board**: a checkerboard from one of the checkers sets, with a strip of masking tape (or a heavy pencil line) laid across it between the fourth and fifth rows: that is the river. Use checkers as soldiers — they are visible from a few feet away. Ignore the colors of the squares; every square is a square. Keep this board free for the share.
- **Paper and pencils** for recording hops (grades 2–5 write hops as *from*→*to*, e.g. d3→d5).
- **Ten minutes before**: each adult plays through two rows north (d3→d5, f4→d4, d4→d6) and the T army (Section 5) once with real pennies. Set out pennies and grids. The 4–5 adult may also want to have run through the 19 hops of the 20-soldier army (the Appendix); it takes about five minutes with the list in hand.

### The rule, with its edge cases

Children will probe the rule. The answers:

- Hops go up, down, left or right only. No diagonal hops. A soldier hops over exactly one soldier and lands on the very next square, which must be empty; it cannot hop over two, and cannot hop over an empty square.
- A hop may go in any direction, including south, back toward the camp. (Southward hops are legal and occasionally tempting; they are never useful for going north.)
- The river is a line between two rows, not a wall: hopping across it is the whole point. A soldier in the camp may hop within the camp; a soldier north of the river may hop anywhere.
- One hop at a time; “chain jumps” as in checkers are just several hops, each removing a soldier.
- The printed board has edges but the real game is played on an endless grid. For the missions this never matters (no useful hop leaves the printed board). For the *strip* puzzles, the strip’s ends are walls: a hop may not leave the strip.
- Nobody “wins”; the table works on one position together. A good habit is turn-taking on the hops: whoever’s turn it is proposes the hop, the others say whether it is legal.

## 3 Free handling (0:00–0:05) and the launch (0:05–0:12)

**Free handling.** Let the children pile pennies onto grids and invent things. If a child asks what the game is, show one hop and say *“that is the only move: hop over a neighbor into an empty square, and the one you hopped over goes home.”* Do not pose the missions yet.

**Launch script** (organizer, everyone standing around the checkerboard, seven minutes at most):

1. *“This is a game called Conway’s Soldiers. The checkers are soldiers. A soldier can do one thing.”* Show a hop. Remove the hopped soldier: *“he goes home.”* Show a diagonal hop, a hop over an empty square, and a hop onto an occupied square, and ask the children to shout *no* at each.
2. Point at the tape: *“This is a river. The army camps on the south side. North of the river the land is empty. How far north can one soldier get?”* Put two checkers in a column touching the river and ask a child to get one across. (The back one hops over the front one.) *“One row north: two*

*soldiers. Nobody stays behind."*

3. *"Two rows north? Three? A hundred?"* Collect a couple of guesses without confirming anything. *"Every table has missions about this. At ten to the hour we come back here and each table shows one thing it found."*
4. Send them to the tables. The K-1 adult may leave a minute early to have the pennies out.

Keep the numbers 4, 8, 20 and the word "impossible" out of the launch; discovering them is the tables' work.

## 4 Table K-1 (two kindergartners, one first grader)

Everything on these pages is read aloud by the adult; the children handle pennies. Expect to spend the whole time on pages 1-4, and treat page 5 (the T) as a joint project if the group is quick. Turn-taking on hops keeps three small children engaged: one hops, the other two check.

### Page 1: The Hop (about 6 minutes)

Demonstrate on the first strip: penny hops over penny, the hopped penny is taken off. Then the  $3 \times 3$  practice square: side by side, then one above the other. Let each child do several hops in each direction; the physical motion (pick up, hop, land, remove) is the learning.

*Can the dark soldier hop?* **A: yes** (over the gray one into the empty square). **B: no** (the next square is empty — nothing to hop over). **C: no** (there is a soldier to hop over, but the landing square is full). **D: no** (slanted). **E: yes** (it hops left over the gray one into the empty square). Children often answer C "yes"; put the pennies down and let them try.

### Page 2: Leave One (about 8 minutes)

Set up each strip from the gray circles; the children hop until one penny remains.

**1** (pennies on squares 1, 2): one hop. **2** (1, 2, 4): two hops; hop 1 over 2 onto 3, then either 3 over 4 onto 5 or 4 over 3 onto 2. Every order works. **3** (1, 3, 4): two hops, but the first hop matters: 4 over 3 onto 2, then 1 over 2 onto 3. If a child first hops 3 over 4 onto 5, the pennies on 1 and 5 are stuck — *"set it up again and try the other hop."* **4** (1, 3, 4, 5, 6 on the six-strip): four hops. The only possible first hop is 4 over 3 onto 2; after that (pennies on 1, 2, 5, 6) every order works.

The counting question: five pennies, one left, so four went home, and every hop sends exactly one home — **four hops, always**. Ask it again for puzzles 2 and 3 (two hops each). The rule "hops = start - end" is the one piece of mathematics every K-1 child can take away.

### Page 3: Across the River (about 6 minutes)

Pennies on c3 and c4; the back one hops over the front one onto the star: c3→c5. Then the follow-up: where else can the two start? **Answer:** in any column, but they must be one above the other with the front one touching the river (rows 3 and 4). Side by side along the river they can only hop sideways; one row lower (rows 2 and 3) the hop lands on row 4, still in the camp, and the lone soldier is stuck. Let them discover the stuck cases; the lesson is *"a soldier needs a friend to hop over, and after the hop the friend is gone."*

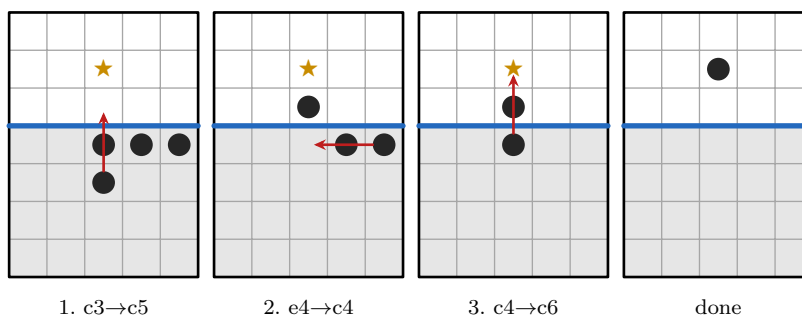
### Page 4: Two Rows North (about 8 minutes)

Pennies on c3, c4, d4, e4; star on c6.

1. c3→c5 (over c4); 2. e4→c4 (over d4); 3. c4→c6 (over c5). — *"up, sideways, up."*

Legal but useless first hops exist (d4→b4 over c4, or c4→c2 down over c3); after either, the position is dead. Do not prevent them — let the table notice it is stuck and reset. If nobody finds the sideways hop

after a few resets, ask *“the soldier on c5 needs a friend right under him. Is there a way to get a soldier onto c4 again?”*

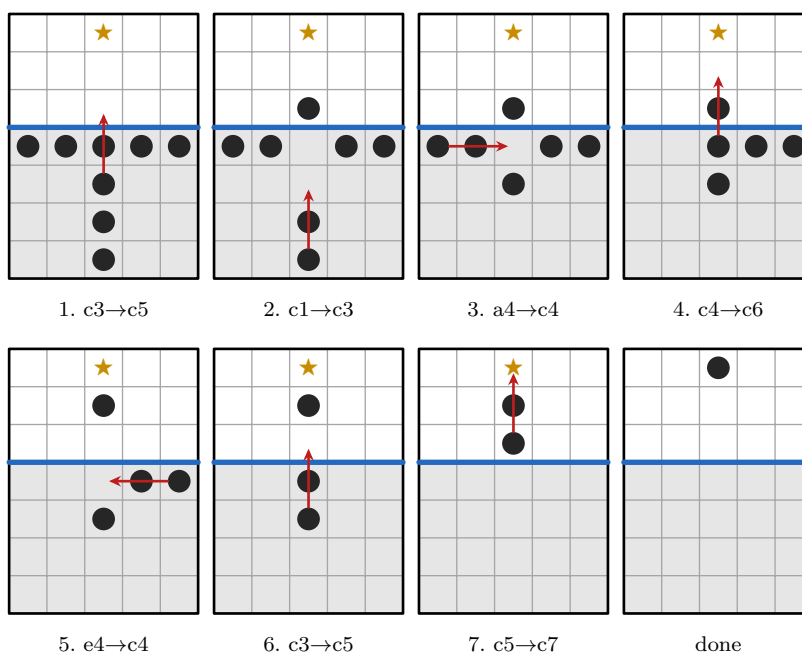


### Page 5: Three Rows North — with the adult (if time)

Eight pennies in a T: a4–e4 and c1, c2, c3; star on c7. This is a seven-hop sequence, too long for K–1 to find alone; run it as a joint project, asking at each step *“which soldier can hop up?”* and letting them execute.

1. c3→c5 (over c4); 2. c1→c3 (over c2); 3. a4→c4 (over b4); 4. c4→c6 (over c5); 5. e4→c4 (over d4);
6. c3→c5 (over c4); 7. c5→c7 (over c6).

Afterwards: eight soldiers, one reached the star, so seven went home — seven hops.



### Page 6: Make Your Own

Open-ended. A useful nudge: *“put pennies in a solid row of three — can you leave one?”* (No: the first hop must be from an end, and it leaves two pennies with a gap.) Any solid row of three or more on a strip gets stuck with at least two left; a row of two leaves one. The square camp is for free play; there are no target answers.

#### If the K–1 table stalls or finishes early

- **The hop itself is hard** (common for kindergartners): stay on pages 1–2 for the whole session. Two children who can hop confidently and count their hops have had a full session.
- **Stuck on page 4:** reset and give the “friend under him” hint above, then, if needed, show the sideways hop and let them do the last one.

- **Finished early:** the T on page 5; then “*can you do two rows north with the pennies on the other side (c3, c4, b4, a4)?*” (yes, the mirror image:  $c3 \rightarrow c5$ ,  $a4 \rightarrow c4$ ,  $c4 \rightarrow c6$ ); then “*can three soldiers get two rows north?*” and let them try — they will find it cannot be done, which is true (Section 8).
- **For the share:** the K–1 table shows two rows north with four checkers on the launch board, one child per hop, the adult narrating “*up, sideways, up.*”

## 5 Table grades 2–3 (four third graders)

The center of this table's hour is Mission 3, the eight-soldier T army; budget fifteen minutes for it and hold the hints back as long as the children are still generating ideas. Everything before it should move briskly.

### Page 1: The Hop and the strip warm-ups (about 8 minutes)

Check the rule with the five pictures (allowed: straight hop over a neighbor into an empty square; not allowed: slanted, over an empty square, over two at once). Then the strips.

**W1** (pennies on 1, 2, 4 of five squares): 2 hops. 1 over 2 onto 3; then 3 over 4 onto 5, or 4 over 3 onto 2.

**W2** (1, 3, 5, 6 of six): 3 hops. The first hop is forced: 6 over 5 onto 4. Then 4 over 3 onto 2 (good; finish with 1 over 2 onto 3) — but 3 over 4 onto 5 instead leaves 1 and 5 stuck.

**W3** (1, 2, 4, 5 of five): **impossible**; two pennies always remain. Proof, which third graders can do completely: only two first hops exist, 1 over 2 onto 3 and its mirror image 5 over 4 onto 3. After 1 over 2 onto 3 the pennies are on 3, 4, 5; the only legal hop is now 4 over 3 onto 2 (the other candidates land on full squares or off the strip), leaving 2 and 5, which cannot reach each other. The mirror case is the same. So every line of play ends with two pennies.

**W4** (2, 4, 6, 7 of seven): 3 hops. Forced first hop 7 over 6 onto 5; then 5 over 4 onto 3 (good; finish with 2 over 3 onto 4 or 3 over 2 onto 1), while 4 over 5 onto 6 instead leaves 2 and 6 stuck.

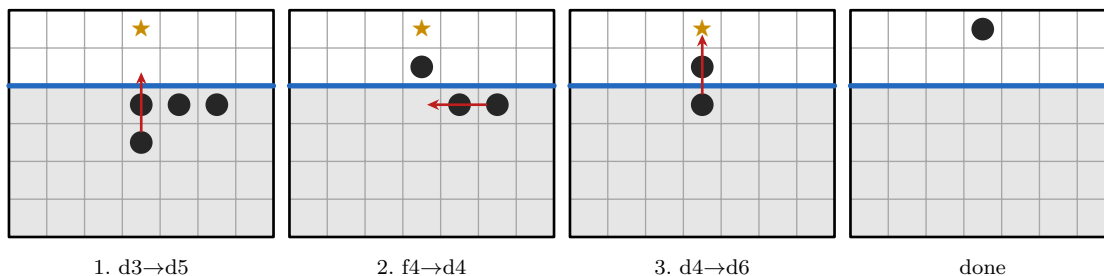
**The counting rule:** every hop removes exactly one soldier, so the number of hops is always (soldiers at the start) – (soldiers at the end); leaving one from four takes exactly three hops, in any order that works. Children who answer “it depends on the order” should count a couple of orders.

### Page 2: Missions 1 and 2 (about 8 minutes)

**Mission 1:** two soldiers in a column touching the river, e.g. d3, d4;  $d3 \rightarrow d5$ . Any column works.

**Mission 2:** the fewest is **four**. With the board's coordinates the two armies that work are the L shapes d3, d4, e4, f4 and its mirror image d3, d4, c4, b4 (plus shifts to other columns). These are the only four-soldier armies; larger armies have more options.

1.  $d3 \rightarrow d5$  (over d4); 2.  $f4 \rightarrow d4$  (over e4); 3.  $d4 \rightarrow d6$  (over d5). (mirror image:  $d3 \rightarrow d5$ ,  $b4 \rightarrow d4$ ,  $d4 \rightarrow d6$ .)

1.  $d3 \rightarrow d5$ 2.  $f4 \rightarrow d4$ 3.  $d4 \rightarrow d6$ 

done

If a group insists three should work, let them try for a few minutes and then send them to the “Why not 3?” question on page 4 (answer below).

### Page 3: Mission 3, three rows north with eight (about 15 minutes)

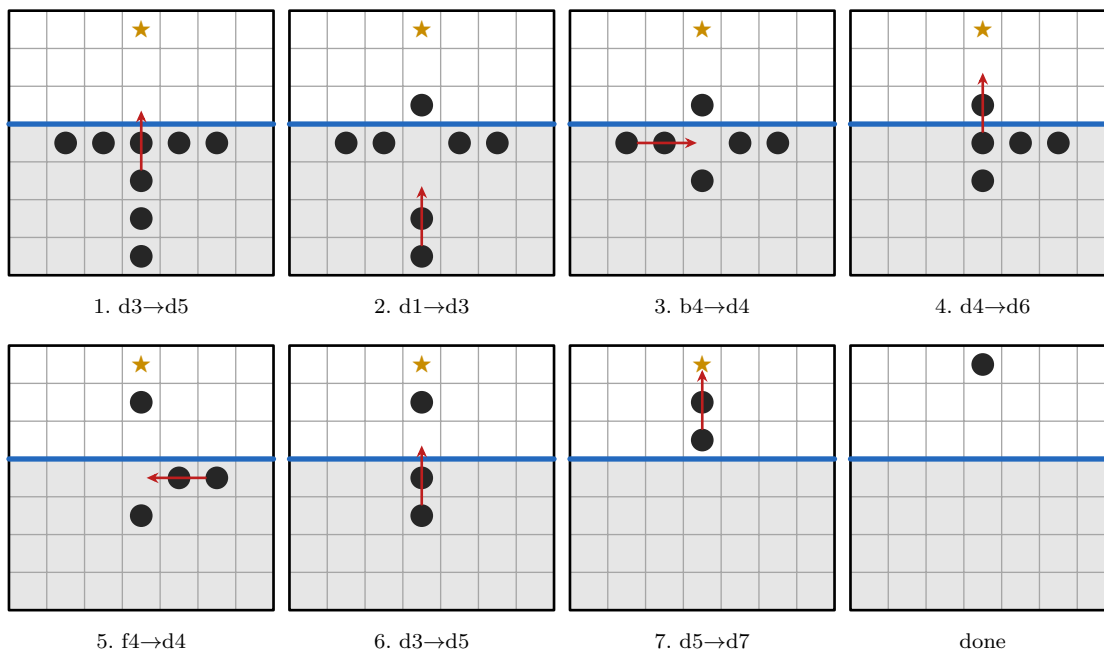
There are exactly three eight-soldier armies that reach d7 (two are mirror images of each other). The nicest is the T:

**The T army:** b4, c4, d4, e4, f4 and d1, d2, d3.

1. d3→d5 (over d4); 2. d1→d3 (over d2); 3. b4→d4 (over c4); 4. d4→d6 (over d5); 5. f4→d4 (over e4); 6. d3→d5 (over d4); 7. d5→d7 (over d6).

**Left-heel army:** b4–f4 and b3, c3, d3. 1. b3→b5 (over b4); 2. c3→c5 (over c4); 3. d3→d5 (over d4); 4. f4→d4 (over e4); 5. d4→d6 (over d5); 6. b5→d5 (over c5); 7. d5→d7 (over d6).

**Right-heel army:** b4–f4 and d3, e3, f3. 1. d3→d5 (over d4); 2. e3→e5 (over e4); 3. f3→f5 (over f4); 4. b4→d4 (over c4); 5. d4→d6 (over d5); 6. f5→d5 (over e5); 7. d5→d7 (over d6).



**Hint ladder** — give one rung at a time, and only when the ideas have dried up, not when the first attempt fails:

- H1. *“Where must the very last hop come from?”* (Straight up: d5→d7 over d6. So just before the end you need soldiers on *both* d5 and d6.)
- H2. *“You already know how to put a soldier on d6 — that is Mission 2. After you do it, what is left on d5?”* (Nothing. Now a *new* soldier has to reach d5.)
- H3. *“To hop onto d5 you need soldiers on d3 and d4 again. Both are empty now. Where could new soldiers come from — from below, from the side?”* (d4 can be refilled sideways, f4→d4 over e4; d3 can be refilled from below, d1→d3 over d2.)
- H4. Show the T and ask them to find the hops. If even that stalls, give hops 1–3 and let them finish.

Watch for a common near-miss: children build the platform on d6, then bring one soldier to d5 by d3→d5, and only then realize d3 needed a friend on d4 that is now gone. That moment is the whole puzzle; let it happen.

### Page 4: Explorer’s page

- **The pattern:** 2, 4, 8 — almost everyone guesses 16 for four rows. The truth is **20**. Tell them at the end, not before they have guessed, and mention that the 4–5 table has the army (the Appendix gives it, with its 19 hops, if they want to try it on a grid sheet).
- **Why not 3?** The soldier that lands on d6 hopped d4→d6 over d5, so just before the last hop

there were soldiers on **d4** and **d5**. The soldier on **d5** got there by **d3**→**d5** over **d4**, which *emptied* **d4**. So **d4** was filled twice: once at the start and once by a hop — and a hop onto **d4** needs two more soldiers. Counting: two soldiers to reach **d5**, two more to refill **d4**, at least four. With only three soldiers there are at most two hops in the whole game: one must put a soldier on row 5 and the other must be the final hop, so the third soldier never moves and must have been standing on **d4** all along — but **d4** was already taken by the soldier that got hopped over. (The other way onto row 6, a sideways hop along row 6, needs two soldiers already on row 6, which is worse.)

- **Another army:** the left-heel and right-heel armies above. Children who find something that looks different have usually shifted the T to another column or mirrored it; check with the list. If they ask whether there are more: no, exactly these three shapes (we checked all possibilities by computer while preparing the packet).
- **How many hops?** Eight soldiers, one reaches the star and none are left over, so seven went home: **seven hops**, for every eight-soldier solution. (Could an eight-soldier solution leave a soldier behind and use six hops? No — the pricing argument of Section 8 shows an eight-soldier army has no soldier to spare.)
- **Four rows north:** the 20-soldier army, the Appendix.

### If the 2–3 table stalls or finishes early

- **Stalled on W3:** do not let the impossible strip eat time; tell them it is impossible and that the proof is “there are only two first hops”, and move on. Come back to it if there is time at the end.
- **Stalled on Mission 3:** the hint ladder, one rung every three or four minutes. Every child should see a soldier reach **d7** at this table today, even if the adult supplies the last hops.
- **Finished Mission 3 by 0:35:** the explorer’s page in order; then the strip question from the 4–5 packet: *“on a one-row strip, with the river between two squares, how far past the river can a soldier get?”* (One square. Two is impossible; they can convince themselves by trying, and Section 8 has the proof if they want to hear it.) Then the 20-soldier army on a grid sheet.
- **For the share:** the T army on the launch board, seven hops, one child per hop.

## 6 Table grades 4–5 (two fourth graders, one fifth grader)

This table gets the real theorem. The hour has three movements: play (rows one to three, fewest soldiers), a first proof (three soldiers cannot make two rows), and the pricing argument on the strip, which is Conway’s proof in one dimension. Page 4 (the 20-soldier army) is for early finishers or for taking home. Aim to start page 3 by 0:35 even if page 2 is not fully written up.

### Page 1: strips (about 8 minutes)

**S1** (pennies on 1, 2, 4, 6 of six squares): 3 hops. First 1 over 2 onto 3 (the only first hop); then 3 over 4 onto 5 (good; finish with 6 over 5 onto 4) — not 4 over 3 onto 2, which leaves 2 and 6 stuck.  
**S2** (2, 4, 6, 7 of seven): 3 hops. Forced 7 over 6 onto 5; then 5 over 4 onto 3; then 2 over 3 onto 4 or 3 over 2 onto 1.  
**S3** (1, 2, 4, 5 of five): **impossible**. Proof by exhaustion: the only first hops are 1 over 2 onto 3 and its mirror image; after 1 over 2 onto 3 (pennies on 3, 4, 5) the only legal hop is 4 over 3 onto 2, leaving 2 and 5 apart. So two pennies always remain.

**How far along a strip?** (a) Soldiers on squares 5 and 6; 5 over 6 onto 7. (b) Square 8 is impossible for every army, however large, with or without gaps — that is the theorem of page 3. Let them try for five minutes and collect the feeling that “every time I get a soldier to 7, square 6 is empty and refilling it costs more than I have.” That is exactly right, and the prices make it precise.

### Page 2: rows one, two, three (about 12 minutes)

Fewest soldiers: **2, 4, 8**. Armies and hops:

**One row:** d3, d4; d3→d5.

**Two rows:** d3, d4, e4, f4 (or its mirror image); 1. d3→d5 (over d4); 2. f4→d4 (over e4); 3. d4→d6 (over d5).

**Three rows, the T army:** b4–f4 and d1, d2, d3; 1. d3→d5 (over d4); 2. d1→d3 (over d2); 3. b4→d4 (over c4); 4. d4→d6 (over d5); 5. f4→d4 (over e4); 6. d3→d5 (over d4); 7. d5→d7 (over d6). (Two other eight-soldier armies: b4–f4 with b3, c3, d3, and its mirror image; hops in Section 5.)

The hint ladder for the T is in Section 5; fourth and fifth graders usually get there with rung H1 alone (*“what must the last hop be?”*) — working backwards from the target is the technique to name out loud, because it is also how the 20-soldier army is solved.

**Prove it: three soldiers cannot make two rows.** Model answer: the soldier that lands on d6 must arrive by d4→d6 over d5 (a sideways hop along row 6 would need two soldiers already on row 6, even harder). So just before the final hop, d4 and d5 are both occupied. The soldier on d5 arrived by d3→d5 over d4 — which removed the soldier on d4. So some hop later refilled d4, and every hop uses two soldiers: two for reaching d5, two for refilling d4, four at least. With three soldiers, cleaner still: there are at most two hops in the whole game; one of them must be the hop onto row 5 and the other the hop onto row 6, so the third soldier never moves; it must be the one on d4 at the end, so it stood on d4 from the beginning — but d4 was occupied by the soldier that was hopped over in the first hop. Contradiction. *How few soldiers for three rows?* At least 8 is best shown with prices (Section 8); offer it to a child who asks.

### Page 3: prices — why square 8 is impossible (about 12 minutes; the adult leads)

Prices by distance from the target square 8: square  $8 - d$  costs  $\sigma^d$  where  $\sigma = 0.618\dots$ ; the page shows them rounded. Walk through the five items with the children doing the arithmetic.

1. **Price rule.**  $.618 = .382 + .236$ ,  $.236 = .146 + .090$ ,  $.146 = .090 + .056$ ,  $.090 = .056 + .034$ ; the rounded glitch is  $.056 \neq .034 + .021 = .055$ . To the right of square 8 the prices mirror ( $.618, .382, \dots$ ), so every square on the endless strip has a price.
2. **Hops never raise the total.** Toward the target: before  $p_k + p_{k+1}$ , after  $p_{k+2}$ , *equal* by the price rule. Away from the target: before  $p_{k+2} + p_{k+1}$ , after  $p_k = p_{k+2} - p_{k+1}$ , smaller by  $2p_{k+1}$ . (The same two computations cover hops to the right of square 8 with the mirrored prices, and a hop across square 8 itself, which loses the middle square's price.)
3. **Running totals:**  $.382, .618, .764, .854, .910, .944, .965, .978, .986, \dots$  — creeping toward 1 and never reaching it. The nice observation: the shortfall after each step is exactly the price of the square just to the right of the last one added ( $1 - .764 = .236$ , the price of square 5, and so on). Exactly: the endless camp is  $\sigma^2 + \sigma^3 + \sigma^4 + \dots = \sigma^2/(1 - \sigma) = \sigma^2/\sigma^2 = 1$ , using  $1 - \sigma = \sigma^2$ .
4. **Punchline.** A finite army leaves out infinitely many camp squares, so it is worth less than \$1; hops never raise the total; a soldier on square 8 alone is worth \$1. So no soldier ever stands on square 8.
5.  $\sigma = (\sqrt{5} - 1)/2 = 0.6180\dots$ , the reciprocal of the golden ratio  $\varphi = 1.6180\dots$ ;  $\sigma^2 = 0.3819\dots$  and  $\sigma + \sigma^2 = 1$  exactly.

Two things the children may ask. *Why these prices?* Because the price rule (a square's price equals the sum of the two beyond it) is exactly what makes a forward hop an even trade; if you start with \$1 on the target and choose any other number for square 7, the rule eventually produces a negative price somewhere to the left. *Why does the army being finite matter?* With the endless army the total is exactly \$1, so a soldier on square 8 would have to be the only soldier left — but finitely many hops remove finitely many soldiers, leaving infinitely many, worth more than \$0. Either way, square 8 is safe.

**Two dimensions.** Same prices by walking (taxicab) distance from a target square five rows north; the endless camp adds up to exactly \$1 again (the calculation is in Section 8), so five rows north is impossible; four rows is done by the army on page 4. The strip is the one-dimensional case of the same theorem: one square on the strip, four rows on the plane.

## Page 4: the 20-soldier army

The army (rows of 7, 6 and 5 soldiers stepping back from the river, plus two feet) and its 19 hops are in the Appendix; the final hop is  $e6 \rightarrow e8$ . The children set it up on the grid sheet. It is a serious puzzle; working backwards from  $e8$  is the method, and the appendix's snapshot pictures let the adult give the next hop whenever the group asks for it. It does not need to be finished today.

### If the 4–5 table stalls or finishes early

- **The T stalls past 0:30:** give the hint ladder quickly, or simply show the T and have them find the hops; page 3 matters more than finding the T unaided.
- **Prices feel abstract:** do item 2 with pennies on the strip and the prices written under the squares: physically trade two soldiers for one and add up before and after. Item 3 with a calculator if one is handy.
- **Finished early:** (i) the 20-soldier army; (ii) *“use prices to show that seven soldiers cannot make three rows”* (Section 8: the seven most valuable squares add up to .944); (iii) *“find the other two eight-soldier armies”*; (iv) *“on the strip, what is the cheapest army that reaches square 7, in dollars?”* (Measured from the new target, square 7, the soldiers on squares 5 and 6 cost  $\sigma^2 + \sigma = .382 + .618 = 1$ , exactly the price of the target; a good moment to notice that prices are always measured from the square you are trying to reach, and that an army worth exactly \$1 has nothing to spare.)
- **For the share:** the table states the theorem — four rows possible, five impossible — and shows the price strip, or the 20-soldier army if they got somewhere with it. One sentence on the idea: *“we gave every square a price so that hops never make the army richer, and the whole endless camp is worth exactly the price of one square in row five.”*

## 7 The share (0:50–1:00)

Everyone around the checkerboard. Three minutes per table, the organizer keeping time.

1. **K–1:** two rows north with four checkers, three children hopping in turn, the adult chanting *“up, sideways, up.”* Ask the room: *“how many soldiers went home?”* (Three.)
2. **Grades 2–3:** the T army and its seven hops. Ask them what the last hop had to be and how they figured out where the eighth soldier goes.
3. **Grades 4–5:** the theorem, in their words: four rows can be done (show the shape of the 20-soldier army on a grid sheet), five rows cannot, ever. One sentence about prices.
4. **Organizer closes:** write 2, 4, 8, 20 on the board and ask what everyone guessed after 2, 4, 8. *“Patterns are hints, not proofs: the fourth number is 20, and the fifth number does not exist. On a strip of one row the limit is one square; on the plane it is four rows. Same game, same proof.”*

## 8 Mathematical background and complete arguments

### Setting

Squares are  $(x, y)$  with integer coordinates; the camp is  $y \leq 0$  (the river runs along the top of row  $y = 0$ ), and “ $r$  rows north” means  $y = r$ . On the printed 7-wide boards the camp is rows 1–4 and the target column is d, so  $(x, y)$  is the square  $x$  columns to the right of column d in row  $4 + y$ ; on the K–1 boards the target column is c; on the 9-wide board for the 20-soldier army it is e. An *army* is any finite set of camp squares. A hop takes soldiers on squares  $A$  and  $B$  (adjacent) and an empty square  $C$  (in line beyond  $B$ ) to a single soldier on  $C$ .

## The hop count

Each hop removes exactly one soldier, so the number of hops in any play equals the number of soldiers at the start minus the number at the end. This is why every 8-soldier solution for three rows takes 7 hops (once the pricing shows no soldier can be spared), and every 20-soldier solution for four rows takes 19.

## Prices (Conway's pagoda function)

Let  $\sigma = (\sqrt{5} - 1)/2 \approx 0.618$ , the positive root of  $\sigma^2 + \sigma = 1$ ; equivalently  $\sigma^d = \sigma^{d+1} + \sigma^{d+2}$  for every  $d$ . Fix a target square  $T$  and give every square  $S$  of the plane the price  $\sigma^{d(S)}$ , where  $d(S)$  is the taxicab distance from  $S$  to  $T$ . The price of a position is the sum of the prices of its occupied squares.

**Lemma (hops never raise the price).** Along any line of three consecutive squares  $A, B, C$  the distances to  $T$  are, in order, either  $(d, d-1, d-2)$ , or  $(d, d+1, d+2)$ , or  $(d, d-1, d)$ ; the last happens when  $B$  lies in  $T$ 's row or column and the hop crosses it. (Along a row or column the taxicab distance is a V-shaped function with its minimum at  $T$ 's row or column, so a three-term stretch of it is monotone or has its minimum in the middle.) A hop from  $A$  over  $B$  to  $C$  changes the price by

$$\sigma^{d-2} - \sigma^d - \sigma^{d-1} = 0, \quad \sigma^{d+2} - \sigma^d - \sigma^{d+1} = -2\sigma^{d+1} < 0, \quad \sigma^d - \sigma^d - \sigma^{d-1} = -\sigma^{d-1} < 0$$

in the three cases. So the price of a position never increases.

**Corollary (a lower bound on armies).** If some army reaches  $T$ , its price is at least 1: at the moment a soldier stands on  $T$  the position is worth at least  $\sigma^0 = 1$ , and the price never went up. Therefore an army that reaches  $T$  must contain squares whose prices add to at least 1.

## The strip: one square past the river, never two

On a one-row strip with the camp being squares  $\leq 0$  and target  $T = 2$ , the camp's total price is

$$\sigma^2 + \sigma^3 + \sigma^4 + \cdots = \frac{\sigma^2}{1 - \sigma} = \frac{\sigma^2}{\sigma^2} = 1.$$

A finite army omits infinitely many camp squares, all of positive price, so its price is strictly less than 1; by the corollary it cannot reach square 2 (and hence nothing beyond it either: the first square beyond 1 that a soldier ever reaches would be square 2, by a hop from square 0 over square 1). Square 1 is reached by soldiers on  $-1, 0$  (price  $\sigma^3 + \sigma^2$  relative to square 2; relative to the target square 1 they are worth  $\sigma^2 + \sigma = 1$ , exactly the price of the target, as they must be). On the student page the river sits between squares 6 and 7, so "square 2" is square 8 and "square 1" is square 7.

## The plane: four rows yes, five rows never

Take  $T$  five rows north, at  $(0, 5)$ . The camp row  $y = -k$  ( $k \geq 0$ ) has the square  $(0, -k)$  at distance  $5 + k$  and, for each  $j \geq 1$ , two squares  $(\pm j, -k)$  at distance  $5 + k + j$ , so its total price is

$$\sigma^{5+k} + 2 \sum_{j \geq 1} \sigma^{5+k+j} = \sigma^{5+k} + \frac{2\sigma^{6+k}}{1 - \sigma} = \sigma^{5+k} + 2\sigma^{4+k}.$$

Summing over  $k \geq 0$ , the entire endless camp is worth

$$\frac{\sigma^5 + 2\sigma^4}{1 - \sigma} = \frac{\sigma^5 + 2\sigma^4}{\sigma^2} = \sigma^3 + 2\sigma^2 = \sigma(1 - \sigma) + 2(1 - \sigma) = 2 - \sigma - \sigma^2 = 1.$$

So a finite army is worth less than 1 and can never put a soldier on  $(0, 5)$ ; by translation the same holds for every square of row 5, and rows beyond 5 cannot be reached without first reaching row 5 (a soldier entering row  $r \geq 5$  for the first time arrives by an upward hop from row  $r-2$  over row  $r-1$ ; if  $r$  were the first row  $\geq 5$  ever entered then  $r-1 \leq 4$ , hence  $r = 5$ ). Even the endless army fails: its price is exactly 1, so a soldier on  $(0, 5)$  would have to be alone on the board, but finitely many hops leave infinitely many soldiers. Rows 1–4 are reachable by the armies of 2, 4, 8, 20 below, so *four rows north is the exact limit*.

## Smallest armies for rows 1 to 4

Prices are now measured from a target  $T$  in row  $r$ ; the camp squares at distance  $d$  from  $T$  number  $1, 3, 5, 7, \dots$  for  $d = r, r+1, r+2, \dots$  (the square directly below  $T$ , then the three squares at distance one more, and so on).

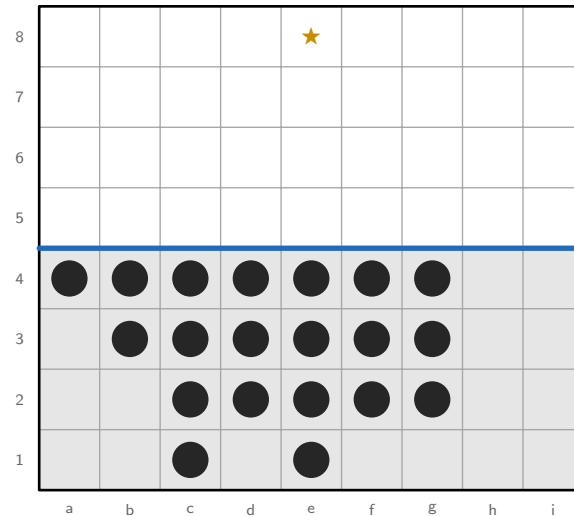
- **Row 1: 2 soldiers.** Two are obviously necessary (a hop needs two) and  $(0, -1), (0, 0)$  suffice.
- **Row 2: 4 soldiers.** With  $T = (0, 2)$  the three most valuable camp squares are worth  $\sigma^2 + 2\sigma^3 \approx .382 + .472 = .854 < 1$ , so three cannot do it (this is the price version of the elementary argument given for the tables). Four suffice:  $(0, -1), (0, 0), (1, 0), (2, 0)$ , worth  $\sigma^3 + \sigma^2 + \sigma^3 + \sigma^4 = 1$  exactly, with hops  $(0, -1) \rightarrow (0, 1), (2, 0) \rightarrow (0, 0), (0, 0) \rightarrow (0, 2)$ . A short search shows this L and its mirror image are the only four-soldier armies for a given target.
- **Row 3: 8 soldiers.** With  $T = (0, 3)$  the seven most valuable camp squares are worth  $\sigma^3 + 3\sigma^4 + 3\sigma^5 \approx .236 + .438 + .271 = .944 < 1$ , so seven cannot. Eight suffice; the three eight-soldier armies (T, left-heel, right-heel) each have price exactly 1, which is why none of them can spare a soldier and every solution takes exactly 7 hops. Section 5 lists their hops.
- **Row 4: 20 soldiers.** With  $T = (0, 4)$  the sixteen squares within distance 7 are worth  $\sigma^4 + 3\sigma^5 + 5\sigma^6 + 7\sigma^7 \approx .936$ , and three more at distance 8 bring the total to exactly 1; so the price bound only says 19 or more. Nineteen is in fact impossible — shown by computer search (Bell, Hirschberg and Guerrero-García, *The minimum size required of a solitaire army*, INTEGERS 7, 2007), and re-checked by an exhaustive search while this packet was prepared: every 19-soldier army whose price reaches 1 was searched to the end and none reaches row 4. Twenty suffice; the same search found exactly eight 20-soldier armies (four mirror-image pairs) reaching a given target square, all with price exactly 1 and all using 19 forward hops. The one in the Appendix is the most memorable shape.

## Context, for the adults

Conway invented the soldiers problem in 1961; the proof above is in Berlekamp, Conway and Guy, *Winning Ways for Your Mathematical Plays*, vol. 4, chapter 23, and in Aigner's article "Moving into the desert with Fibonacci" (*Mathematics Magazine* 70, 1997), which also treats the one-dimensional strip and the Fibonacci flavor of the price rule. The price function is an example of what the peg-solitaire literature calls a *pagoda function*: any assignment of numbers to the holes of a board such that no jump increases the total. Pagoda functions are the standard certificates of impossibility in peg solitaire (on the classic 33-hole cross board they show, for instance, which single holes can be the last peg's home), and they are the same idea as the monovariants and potential functions of olympiad combinatorics: a quantity that can only go one way, computed at the start and at the supposed end.

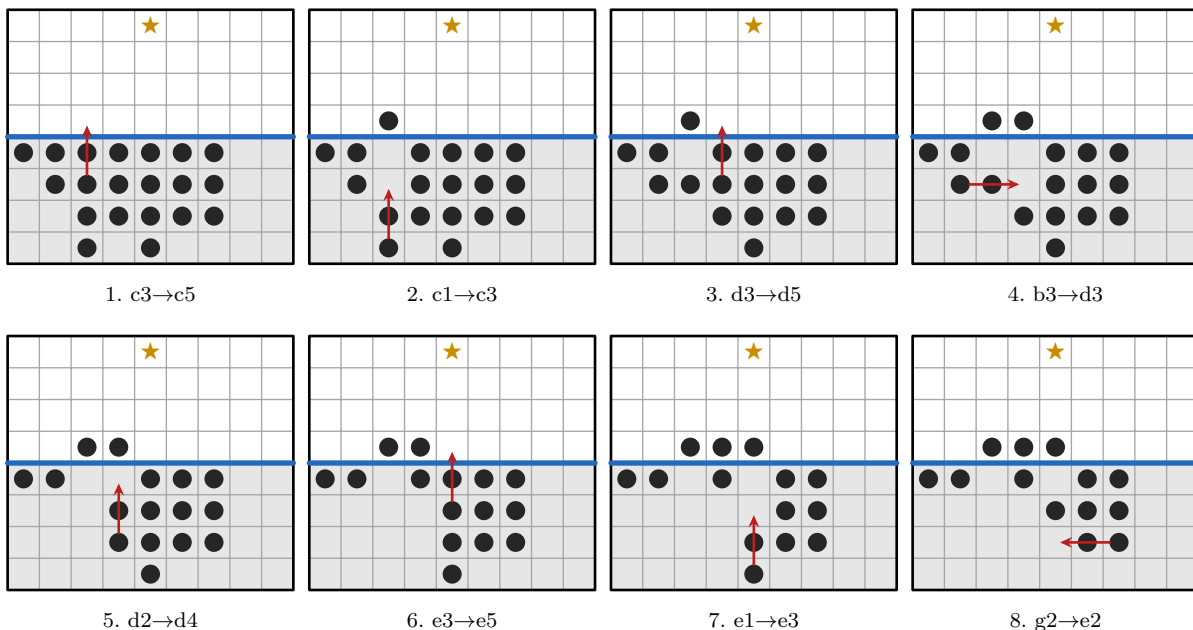
## Appendix: the 20-soldier army for four rows north

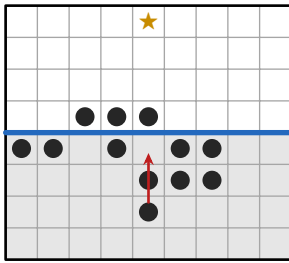
On the  $9 \times 8$  board (columns a–i, camp rows 1–4) the army is a4–g4 (seven), b3–g3 (six), c2–g2 (five), and c1, e1; the target is e8. In  $(x, y)$  coordinates with  $T = (0, 4)$ : row 0 from  $x = -4$  to 2, row -1 from -3 to 2, row -2 from -2 to 2, and  $(-2, -3)$ ,  $(0, -3)$ . Its price is exactly 1; every one of the 19 hops is a forward hop (an even trade), and exactly one soldier is left, on the target.



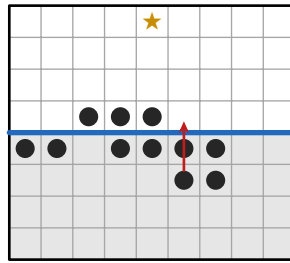
**The 19 hops:** 1. c3→c5 (over c4); 2. c1→c3 (over c2); 3. d3→d5 (over d4); 4. b3→d3 (over c3); 5. d2→d4 (over d3); 6. e3→e5 (over e4); 7. e1→e3 (over e2); 8. g2→e2 (over f2); 9. e2→e4 (over e3); 10. f3→f5 (over f4); 11. g3→g5 (over g4); 12. a4→c4 (over b4); 13. c4→c6 (over c5); 14. d4→d6 (over d5); 15. e4→e6 (over e5); 16. g5→e5 (over f5); 17. e5→e7 (over e6); 18. c6→e6 (over d6); 19. e6→e8 (over e7).

**How to think about it.** The last hop is e6→e8 over e7; e7 is filled by e5→e7 (hop 17), which empties e6, so e6 is refilled sideways by c6→e6 (hop 18). Everything before that builds the soldiers on c6, d6, e6 and e5. Hops 1–11 push five soldiers onto row 5 (after hop 11, c5–g5 are all occupied and a4, b4, d4, e4 still stand in the camp; the camp squares vacated on the way were refilled by sideways and upward hops). Hops 12–15 lift the soldiers of columns c, d, e to row 6, and hop 16 brings g5 to e5. Snapshots before each hop:

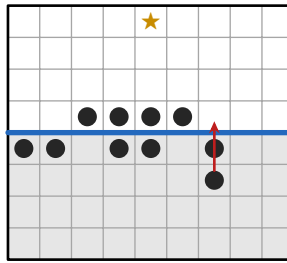




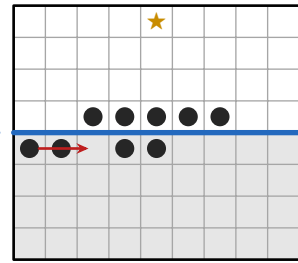
9. e2→e4



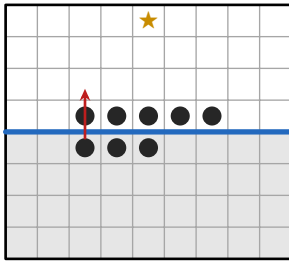
10. f3→f5



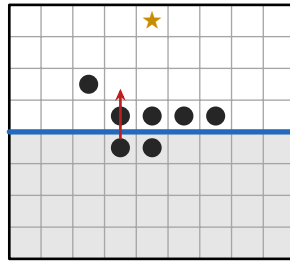
11. g3→g5



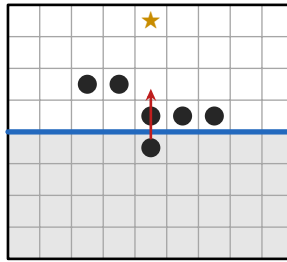
12. a4→c4



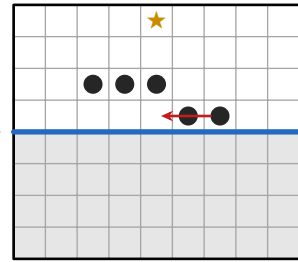
13. c4→c6



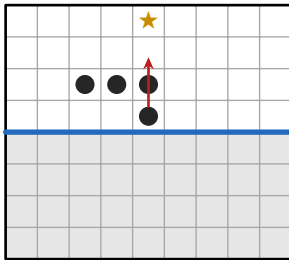
14. d4→d6



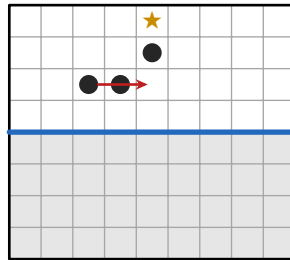
15. e4→e6



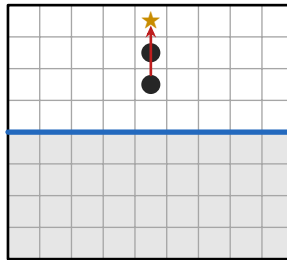
16. g5→e5



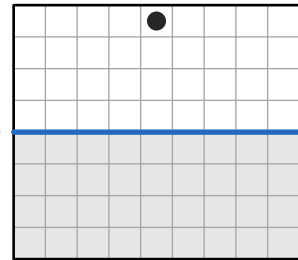
17. e5→e7



18. c6→e6



19. e6→e8



done